If, after inserting those 5 elements above, we call `top`, the element returned will be **“UI breaks on browser X”** and the size of the queue will become 4. If we call `top` again, the next element will be **“CSS style causes misalignment”** and the size will become 3.

As long as the priority queue is implemented correctly, and given the priorities in our examples, we can be sure those two elements will be the ones returned first, independently on the order in which they are inserted.

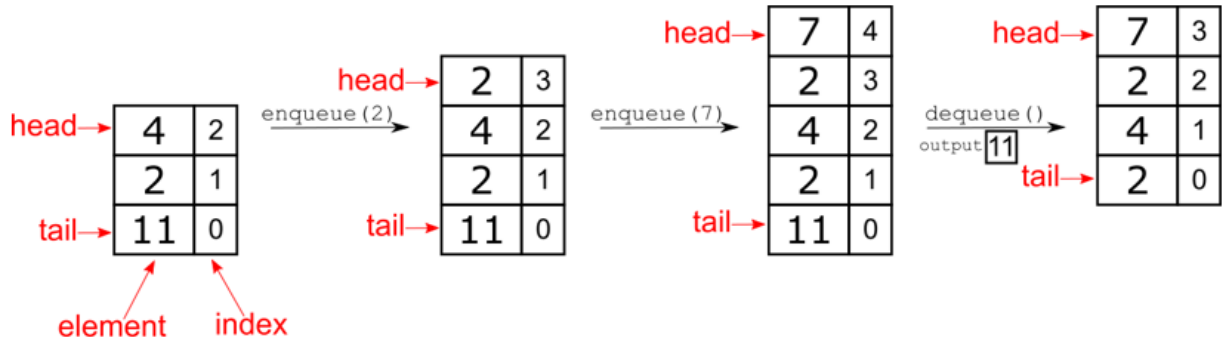


Figure 2.2 Operations on a queue. Insertion (`enqueue`) adds an element to the front of the queue. Delete (`dequeue`) remove the last element in the queue and returns it. With some caution, both operations can be performed in constant time.

2.4.2 Priority Matters: Generalize FIFO

The natural ordering given by how much time an element waits in a line can often be considered the fairest. Sometimes, however, there is something special about some elements that might suggest they should be served sooner